Exercise 9.4

Let A be a regression algorithm that takes an ordered dataset D of size m and returns $\hat{f}(\cdot; D)$, without assuming symmetry. Let $\xi \sim \mathrm{Unif}[0,1]$ and choose a map $\xi \mapsto \sigma_\xi \in \mathfrak{S}_m$ such that σ_ξ is uniform among all permutations. Define

$$s((x,y); D, \xi) = |y - \hat{f}(x; D_{\sigma_\xi})|.$$

Prove that s is a symmetric randomized score function in the sense of Definition 9.1.

Solution Fix any dataset $D = (z_1, \ldots, z_m)$ and any permutation $\pi \in \mathfrak{S}_m$, where $z_j = (x_j, y_j)$. If $\sigma \in \mathfrak{S}_m$, then

$$(D_\pi)_\sigma = (z_{\pi(\sigma(1))}, \ldots, z_{\pi(\sigma(m))}) = D_{\pi\circ\sigma}.$$

The map $\sigma \mapsto \pi\circ\sigma$ is a bijection of $\mathfrak{S}_m$, with inverse $\rho \mapsto \pi^{-1}\circ\rho$. Thus composing a uniform random permutation with the fixed permutation π preserves its uniform distribution:

$$\pi \circ \sigma_\xi \overset{d}{=} \sigma_\xi.$$

In particular, shuffling D_π uniformly has the same distribution as shuffling D uniformly.

To verify the full requirement of Definition 9.1, fix any finite list of evaluation points $(x'_1, y'_1), \ldots, (x'_r, y'_r)$ and define

$$F(\rho) = \big(|y'_j - \hat{f}(x'_j; D_\rho)|\big)_{j=1}^r, \qquad \rho \in \mathfrak{S}_m.$$

Using the same random seed for all r scores, we obtain

$$\begin{aligned}\big(s((x'_j, y'_j); D_\pi, \xi)\big)_{j=1}^r &= F(\pi \circ \sigma_\xi) \\ &\overset{d}{=} F(\sigma_\xi) \\ &= \big(s((x'_j, y'_j); D, \xi)\big)_{j=1}^r.\end{aligned}$$

This proves equality in distribution of the entire score vector for arbitrary evaluation points and any permutation of D, exactly as required by Definition 9.1. No symmetry of A is needed: the random shuffle gives every possible ordering of the data the same probability. ////